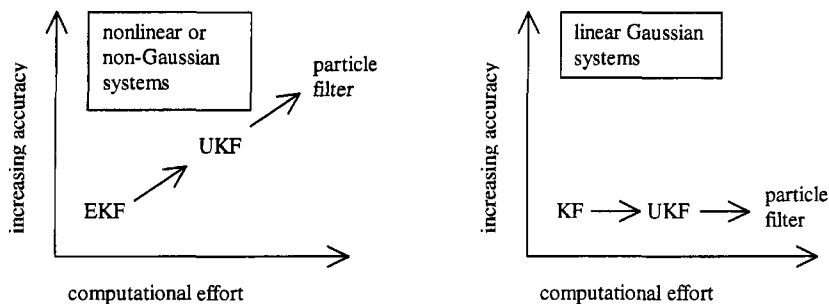


15.4 SUMMARY

In this chapter, we laid the foundation of Bayesian state estimation, and from there we developed the particle filter. In a linear system with Gaussian noise, the Kalman filter is optimal. In a system that is nonlinear, the Kalman filter can be used for state estimation, but the particle filter may give better results at the price of additional computational effort. In a system that has non-Gaussian noise, the Kalman filter is the optimal linear filter, but again the particle filter may perform better. The unscented Kalman filter provides a balance between the low computational effort of the Kalman filter and the high performance of the particle filter. This is depicted in Figure 15.7.



(a) The above figure depicts the increasing computational effort and increasing accuracy that is obtained by going from an EKF to a UKF to a particle filter. This applies to systems that are nonlinear or non-Gaussian.

(b) The above figure depicts the fact that the Kalman filter is optimal for linear Gaussian systems. Going from a Kalman filter to a UKF to a particle filter will increase computational effort but will not improve estimation accuracy.

Figure 15.7 State estimation trade-offs.

The particle filter has some similarities with the UKF (see Chapter 14) in that it transforms a set of points via known nonlinear equations and combines the results to estimate the mean and covariance of the state. However, in the particle filter the points are chosen randomly, whereas in the UKF the points are chosen on the basis of a specific algorithm. Because of this, the number of points used in a particle filter generally needs to be much greater than the number of points in a UKF. Another difference between the two filters is that the estimation error in a UKF does not converge to zero in any sense, but the estimation error in a particle filter does converge to zero as the number of particles (and hence the computational effort) approaches infinity.

Particle filters have found application in a wide variety of areas, including tracking problems [Ris04], demodulation of communication signals [Dou01, Chapter 4], estimation of ecological parameters and populations [Dou01, Chapter 5], image processing [Dou01, Chapter 16], neural network training [Dou01, Chapter 17], fault detection [deF02], speech recognition [Ver02], and pattern recognition [Dou01, Chapter 26]. Particle filtering is a growing area of research with many unexplored avenues and applications. Some of the more important areas of open research include the avoidance of sample impoverishment, methods for determining how many particles are required for a given problem, convergence results [Cri02], application to control and parameter estimation [Mor03, And04], connections with genetic algo-

rithms [Dou01, Chapter 20], real-time implementation issues [Kwo04], and hardware implementations of parallel particle filters (e.g., in field programmable gate arrays).

PROBLEMS

Written exercises

15.1 Consider the scalar system

$$\begin{aligned}x_{k+1} &= x_k + w_k, & w_k &\sim U(-1, 1) \\ y_k &= x_k + v_k, & v_k &\sim U(-1, 1)\end{aligned}$$

where $x_0 \sim U(-1, 1)$. Suppose that the first measurement $y_1 = 1$.

- Use the recursive Bayesian state estimator to find $\text{pdf}(x_1|Y_0)$ and $\text{pdf}(x_1|Y_1)$.
- What is the Kalman filter estimate $\hat{x}_1^+$? How is $\hat{x}_1^+$ related to $\text{pdf}(x_1|Y_1)$?

15.2 Suppose the pdf of an RV x is given as

$$\text{pdf}(x) = \begin{cases} 1 - x/2 & x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

The value of x can be estimated several ways.

- The maximum-likelihood estimate is written as $\hat{x} = \text{argmax}_x \text{pdf}(x)$. Find the maximum-likelihood estimate of x .
- The min-max estimate of x is that value of $\hat{x}$ that minimizes the magnitude of the maximum estimation error. Find the min-max estimate of x .
- The minimum mean square estimate of x is that value of $\hat{x}$ that minimizes $E[(x - \hat{x})^2]$. Find the minimum mean square estimate of x .
- The expected value estimate of x is given as $\hat{x} = E(x)$. Find $E(x)$.

15.3 Suppose you have a measurement $y_k = x_k^2 + v_k$, where v_k has a triangular pdf that is given as

$$\text{pdf}(v_k) = \begin{cases} 1/2 + v_k/4 & v_k \in [-2, 0] \\ 1/2 - v_k/4 & v_k \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

Suppose that five *a priori* particles $x_{k,i}^-$ are given as $-2, -1, 0, 1$, and 2 , and that the measurement is obtained as $y_k = 1$. What are the normalized likelihoods q_i of each *a priori* particle $x_{k,i}^-$?

15.4 Suppose you have a measurement $y_k = v_k/x_k$, where $v_k \sim N(9, 1)$. Suppose that five *a priori* particles $x_{k,i}^-$ are given as $0.8, 0.9, 1.0, 1.1$, and 1.2 , and that the measurement is obtained as $y_k = 10$. What are the relative likelihoods q_i of each *a priori* particle $x_{k,i}^-$?

15.5 Suppose that five *a priori* particles are found to have probabilities $0.1, 0.1, 0.1, 0.2$, and 0.5 . The particles are resampled with the basic strategy depicted in Equation (15.20).

- a) What is the probability that the first particle will be chosen as an *a posteriori* particle at least once?
- b) What is the probability that the fifth particle will be chosen as an *a posteriori* particle at least once?
- c) What is the probability that the five *a posteriori* particles will be equal to the five *a priori* particles (disregarding order)?

15.6 Suppose you have the five particles $x_{k,i}^+ = \{1, 2, 3, -2, 6\}$. What would you propose to use for the estimate of x_k ? What would you estimate as the variance of $\hat{x}_k$?

15.7 Suppose that you have five particles $-1, -1, 0, 1$, and 1 . You want to use the roughening procedure of Section 15.3.1.1 to add a uniform random variable with a variance of $KMN^{-1/n}$ to each particle. What range of K will give a probability of at least $1/8$ that at least one of the roughened particles is less than -2 ?

15.8 Suppose you have the system equation $x_{k+1} = x_k$ and the measurement equation $y_k = x_k^2 + v_k$, where v_k has a triangular pdf that is given as

$$\text{pdf}(v_k) = \begin{cases} 1/2 + v_k/4 & v_k \in [-2, 0] \\ 1/2 - v_k/4 & v_k \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

Suppose that five *a posteriori* particles $x_{k-1,i}^+$ are given as $-2, -1, 0, 1$, and 2 , and that the measurement is obtained as $y_k = 1$. You want to use prior editing to ensure that the -2 particle has at least a 10% chance (after one roughening step) of being selected as an *a posteriori* particle at the next time step. What value of K should you use in your roughening step?

15.9 Suppose you have two particles -1 and $+1$, both with *a priori* probabilities $1/2$. Use the kernel bandwidth $h = 1$ with the regularized particle filter to find the pdf approximations $\hat{p}(x_k = -2|y_k)$, $\hat{p}(x_k = -1|y_k)$, $\hat{p}(x_k = 0|y_k)$, $\hat{p}(x_k = 1|y_k)$, and $\hat{p}(x_k = 2|y_k)$. For what values of x_k is the pdf approximation $\hat{p}(x_k|y_k)$ equal to zero?

15.10 Suppose you have N resampling probabilities q_i with sample mean μ and sample variance S . What is the sample mean and variance of the auxiliary probabilities given by Equation (15.44)?

Computer exercises

15.11 Plot the volume of the n -dimensional unit hypersphere as a function of n for $n \in [1, 20]$.

15.12 Consider two particles $x_1 = 1$ and $x_2 = 2$, with equal probabilities. Generate the approximate pdf using the Epanechnikov kernel with bandwidth $h = h^*$. Generate two separate plots (on the same figure) of the two individual terms in the summation of Equation (15.29), and also generate a plot (on the same figure) of their sum. Repeat for three particles $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$ with equal probabilities.

15.13 Repeat Problem 15.12 with $h = h^*/2$ and with $h = 2h^*$. This shows that the bandwidth selection can have a strong effect on the pdf approximation.

15.14 Kernels other than the Epanechnikov kernel can also be used for pdf approximation [Sim98, Dev01]. Some of the more popular kernels can be described in one dimension as follows.

$$\begin{aligned} \text{Epanechnikov: } K(x) &= \begin{cases} \frac{3}{4}(1-x^2) & |x| < 1 \\ 0 & \text{otherwise} \end{cases} \\ \text{Gaussian: } K(x) &= (2\pi)^{-1/2} \exp(-x^2/2) \\ \text{Uniform: } K(x) &= \begin{cases} \frac{1}{2} & |x| < 1 \\ 0 & \text{otherwise} \end{cases} \\ \text{Triangular: } K(x) &= \begin{cases} 1-x^2 & |x| < 1 \\ 0 & \text{otherwise} \end{cases} \\ \text{Biweight: } K(x) &= \begin{cases} \frac{15}{16}(1-x^2)^2 & |x| < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Bandwidth selection is another matter, but for this problem you can simply use the optimal Epanechnikov bandwidth for all of the kernels.

- a) Repeat Problem 15.12 using Gaussian kernels.
- b) Repeat Problem 15.12 using uniform kernels.
- c) Repeat Problem 15.12 using triangular kernels.
- d) Repeat Problem 15.12 using biweight kernels.

15.15 In this problem, we will explore the performance of the EKF and the particle filter for the system described in Example 15.1.

- a) Run 100 simulations of the EKF and the particle filter with $N = 10$, $N = 100$, and $N = 1000$. What is the average RMS state-estimation error for each case?
- b) Run 100 simulations of the EKF and the particle filter with $N = 100$ using $Q = 0.1$, $Q = 1$, and $Q = 10$. What is the average RMS state-estimation error for each case?

